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Editors-in-Chief
npj Urban Sustainability

Dear Editors,

We are pleased to submit our manuscript entitled "Physics-guided deep learning resolves five-metre urban thermal stress for climate-just delivery routing" for consideration as a Research Article in *npj Urban Sustainability*. All co-authors confirm that this work is original, has not been published or submitted elsewhere, and that no competing interests exist.

Extreme urban heat disproportionately threatens outdoor workers, yet mesoscale weather models cannot resolve the street-level thermal heterogeneity that governs individual exposure. This spatial mismatch is acute for platform-based delivery couriers, whose occupational heat burden is systematically externalised by efficiency-centric dispatch algorithms. We developed HyperClim-Former, a physics-guided deep learning framework that couples mesoscale atmospheric forcings with three-dimensional urban morphology to reconstruct Universal Thermal Climate Index (UTCI) fields at 5-metre, 10-minute resolution. Validated against mobile street-level measurements in Wuhan, China, the framework achieves a root-mean-square error of 1.00 °C, substantially outperforming recent urban microclimate emulators. By coupling these thermal fields with million-scale agent-based courier simulations, we report three principal findings:

(1) Explainable AI attribution identifies dynamic shadow as the dominant driver of daytime UTCI variation (55–60% of predictive influence), with 3D vertical structures yielding monotonic cooling while 2D planar features exhibit non-linear saturation.

(2) Peak delivery demand coincides with the diurnal thermal maximum, confining couriers to severe heat stress (UTCI $\geq$ 32 °C) for over 99% of their working hours.

(3) A Pareto-optimal routing algorithm identifies a systemic sweet spot at which a 1-minute per-order detour budget reduces population-median cumulative heat stress by 21.1% ($p < 0.001$), adding only 4.3 minutes of average delay per shift.

These results align with the scope of *npj Urban Sustainability* on urban climate resilience and social equity. Our framework integrates physical downscaling, agent-based mobility modelling, and multi-objective routing optimisation to establish a transferable paradigm for climate-just urban logistics. The resulting Rider Heat Chokepoint Index offers a spatially explicit planning tool for deploying shading infrastructure at critical exposure hotspots. By showing that minor algorithmic adjustments can yield substantial occupational health benefits, this work demonstrates how digital platform infrastructure can contribute to healthier, more equitable cities.

Thank you for taking the time to consider our manuscript. We would be grateful for the opportunity to contribute to *npj Urban Sustainability* and sincerely welcome any comments or suggestions from the editors and reviewers.

With kind regards,

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